

Center, spread, and distributions

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HES 3000-002, Sport Performance Analytics

WHAT IS “TYPICAL”?

Measures of central tendency

- A method of describing a data set with a single figure
- Main methods:
 - Mean
 - Median
 - Mode
- Most appropriate method depends on the data and question
 - Scale of measurement
 - Outliers and data distribution

Mean, Median and Mode

- Mean: arithmetic mean
 - Determined by the sum of the values in the distribution divided by the number of values.
 - Misleading if outliers or asymmetries are present
- Median: the middle value
 - the data point below which half of the values lie
 - Ignores magnitude, focused on order
- Mode: most frequent value
 - Rarely used for interval or ratio data

Mean, Median and Mode

- Mean: arithmetic mean
 - =AVERAGE(range)
- Median: the middle value
 - =MEDIAN(range)
- Mode: most frequent value
 - =MODE.SNGL(range)

Variation and distribution

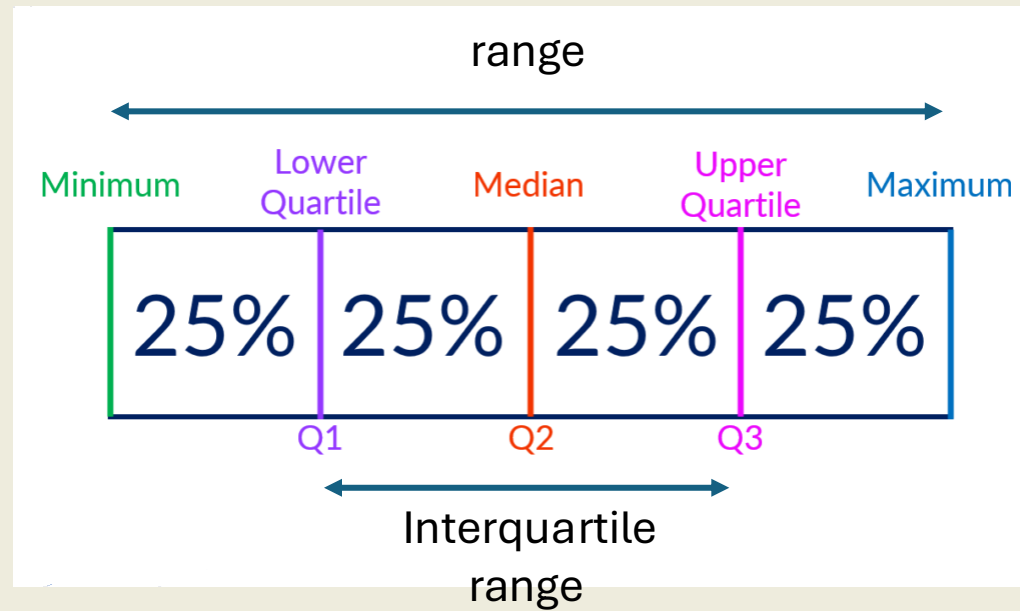
Measures of variability

- The degree of spread or distribution of the data.
- Can be used to show consistency or meaningful differences.
- Methods:
 - Range
 - IQR
 - Variance
 - Standard Deviation

Range

- Range: distance between the smallest and largest values.
 - Sensitive to outliers
- IQR: spread of the middle 50% of the data.
 - Less sensitive to outliers
- Standard Deviation: square root of variance
 - Typical deviation from the mean
 - Same units as the data

Range and IQR

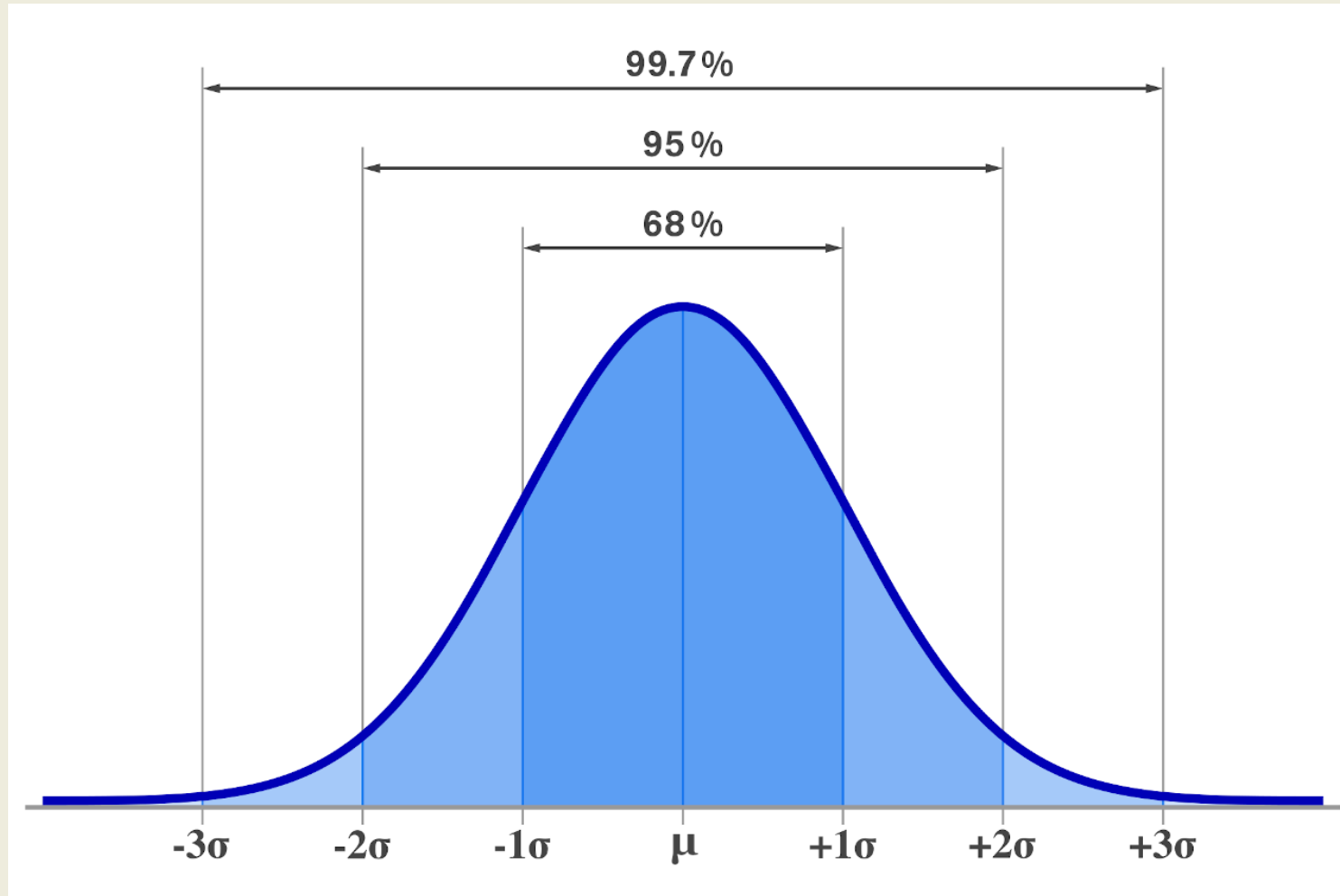


Standard Deviation

- Range: distance between the smallest and largest values.
 - Sensitive to outliers
- IQR: spread of the middle 50% of the data.
 - Removes outliers
- Standard Deviation: Typical deviation from the mean
 - square root of variance
 - Same units as the data

Standard Deviation

Deviation

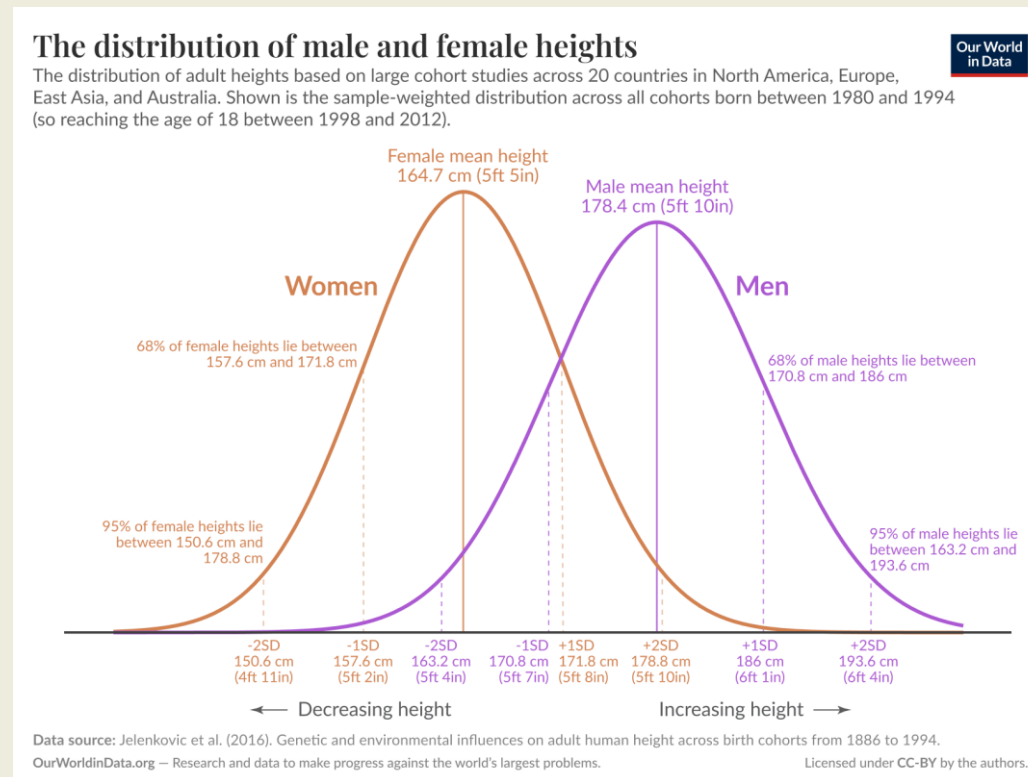


mean

Distribution in data

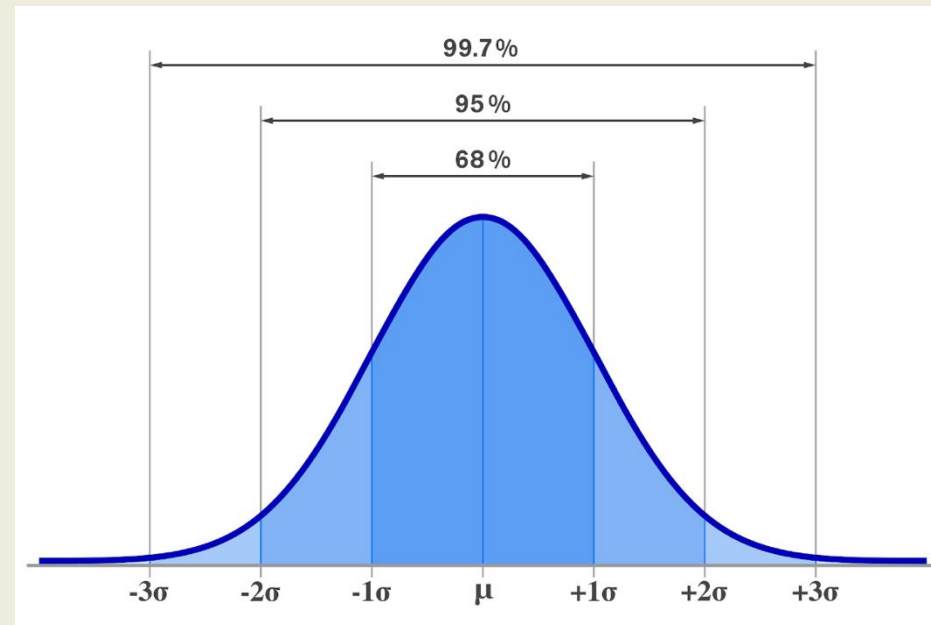
Understanding Distributions

- The Normal Distribution (Bell Curve):
 - Many biological and physical traits roughly approximate a normal distribution



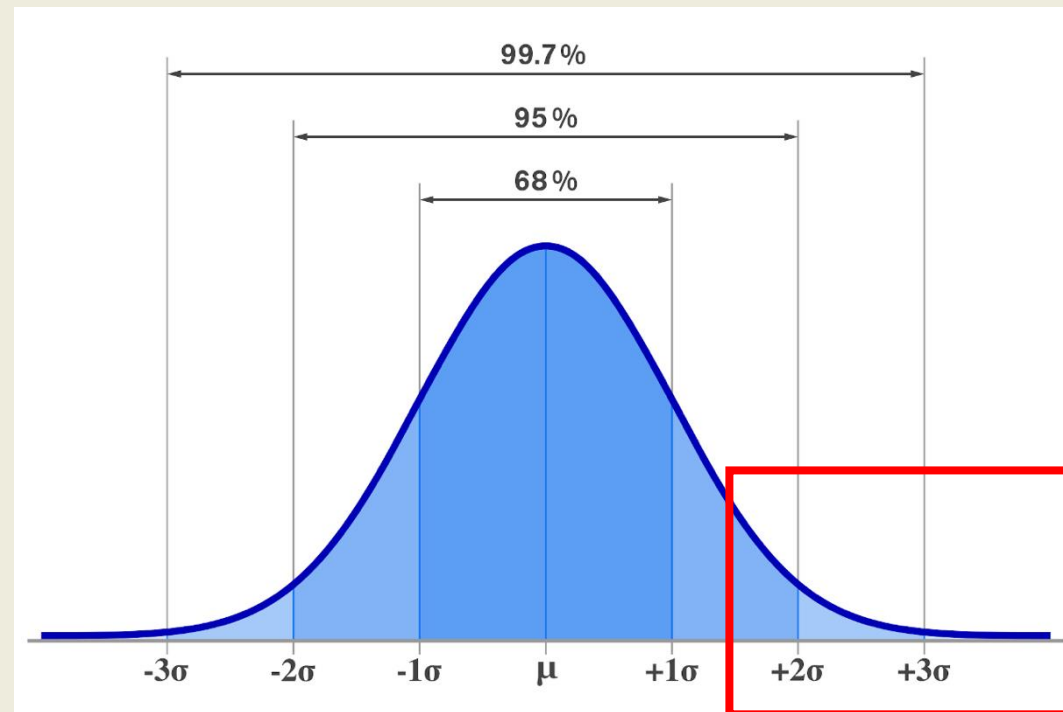
Understanding Distributions

- The Normal Distribution (Bell Curve):
 - Many biological and physical traits roughly approximate a normal distribution
 - Defined by mean and standard deviation (symmetric).



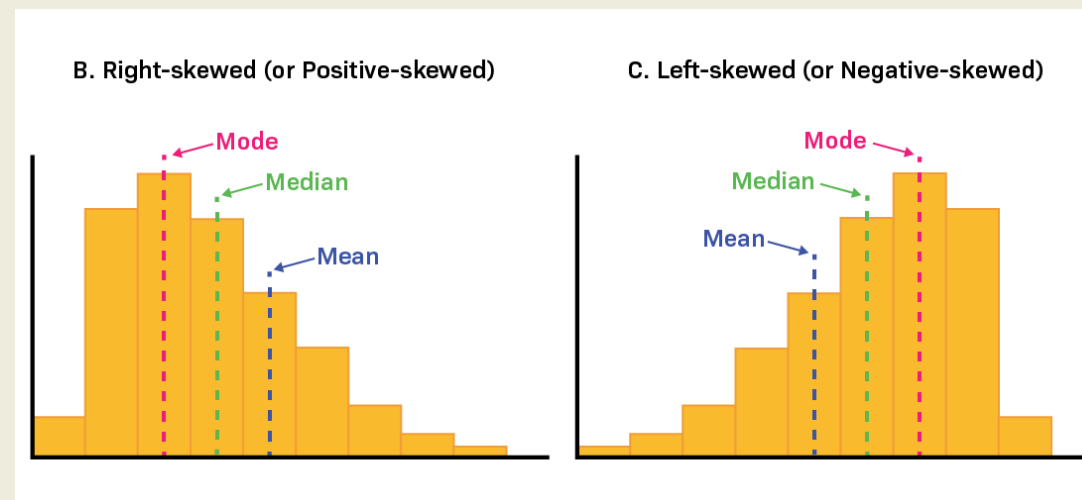
Distribution in Sports

- Data in sports often not-normal



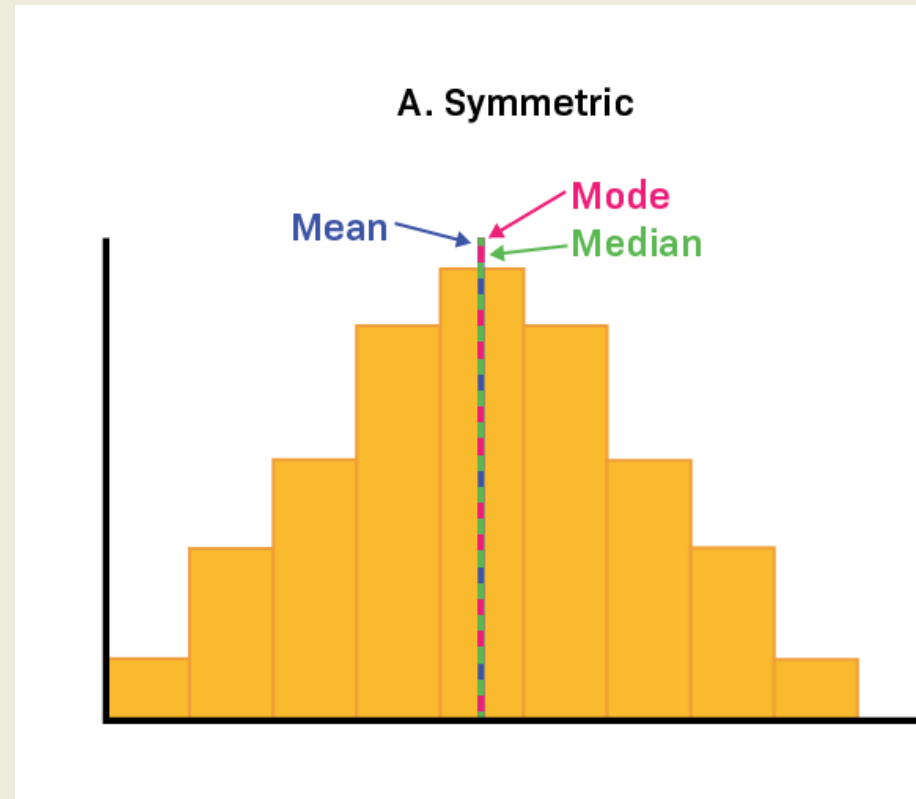
Skew in Data

- Right-Skewed (Positive Skew):
 - Most players have low counts, but a few superstars skew the tail high.
- Left-Skewed (Negative Skew):
 - Values clustered near the maximum ceiling, with a few poor performances trailing off.

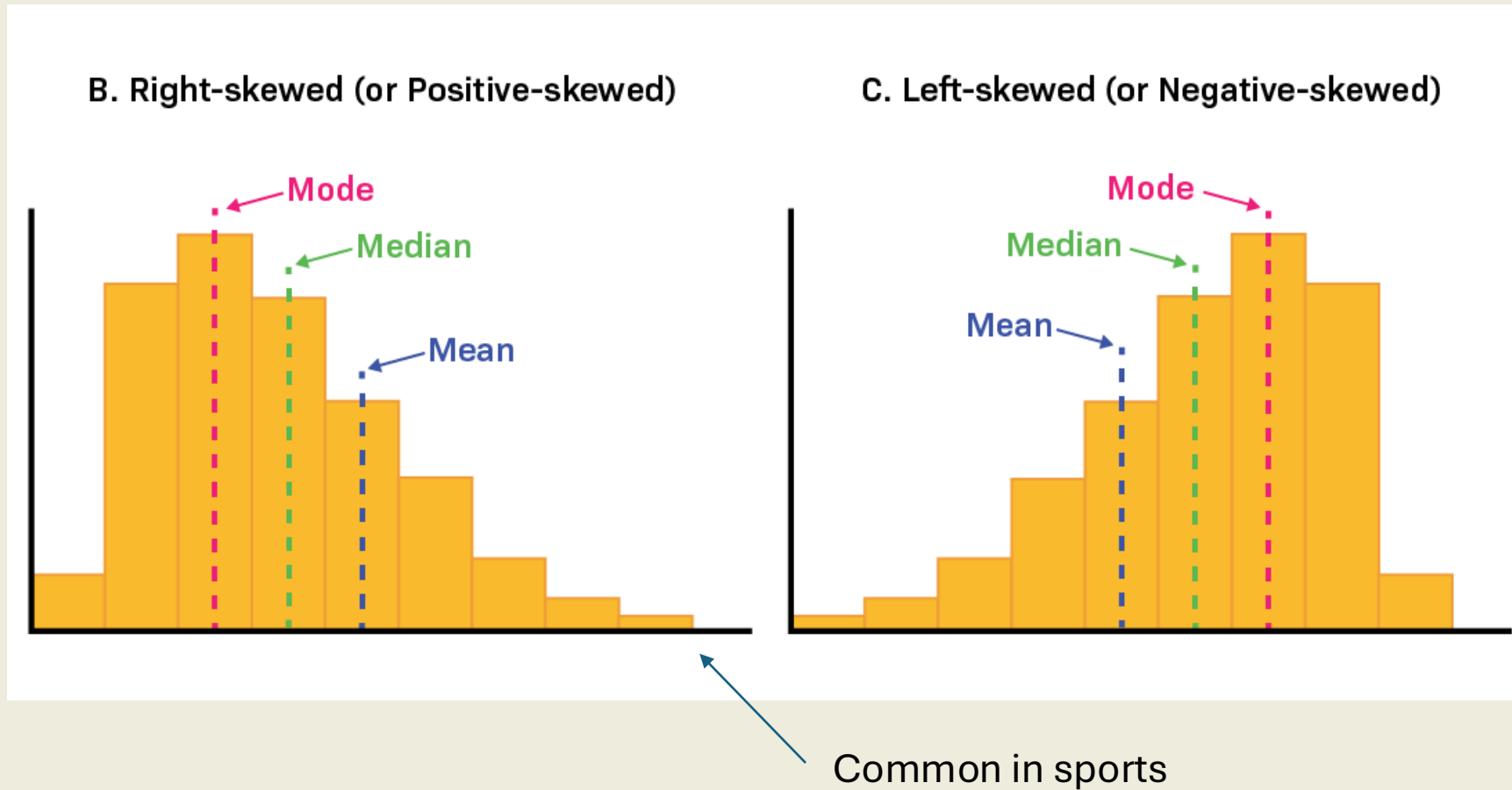


Measuring center in non-normal data

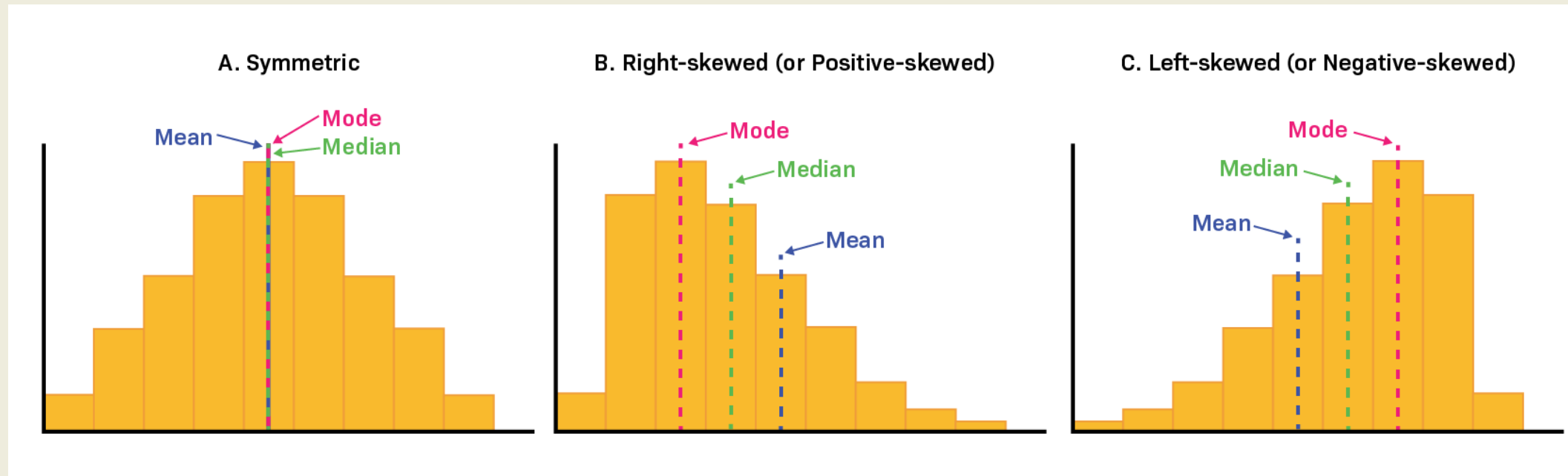
The center in symmetric data



The center in asymmetric data



Does distribution change appropriate method?



Grouped summary

Grouped summary statistics

Why Segment Your Data?

- Looking at team / league average often masks the truth because different positions have entirely different physical and tactical requirements.
- Example:
 - Lumping center-backs and wingers together in soccer makes total high-speed distance look "average."
 - Segmenting by position reveals that wingers log massive high-speed workloads, while central defenders rely more on mechanical impacts and aerial duels.

Simpson's paradox

Who is the better shooter?

	2pt made	2pt att	3pt made	3pt att
Player A	12	16	1	4
Player B	4	4	8	16

	2pt%	3pt%
Player A	75%	25%
Player B	100%	50%

Simpson's paradox

Who is the better shooter?

- $FG\% = (2pt\ made + 3pts\ made) \div (2pt\ attempted + 3pts\ attempted)$

	2pt%	3pt%	FG%
Player A	75%	25%	65%
Player B	100%	50%	60%

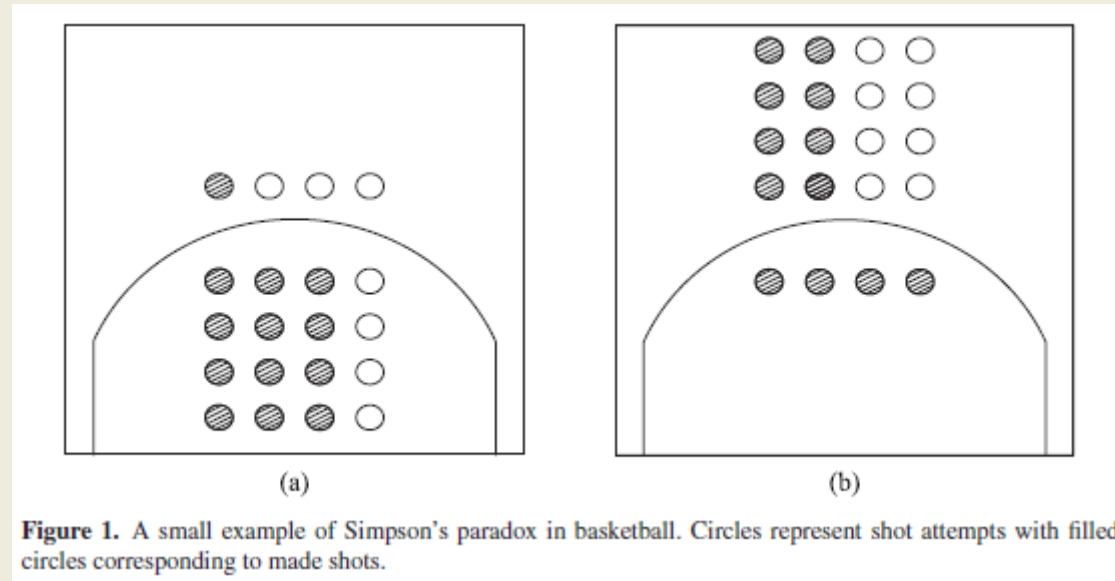
Player B

Player A

How can this be??

Simpson's paradox

Who is the better shooter?



Visualizing Distributions and Groups

- Beyond Summary Tables:
 - Numbers like means and standard deviations tell part of the story, but visual shapes reveal the full picture.
 - Bar graphs, scatter plots and line graphs can show averages and averages over time
 - Histograms, density plots, and boxplots are useful for spotting positional differences and skewness.

Anscombe's quartet

Steph Curry's 2021-2022 NBA Playoff shot locations

- Mean, standard deviation and correlation of shot locations.

mean of x	0.73977
mean of y	14.71046
standard deviation of x	11.26635
standard deviation of y	10.42091
correlation of x and y	-0.00989

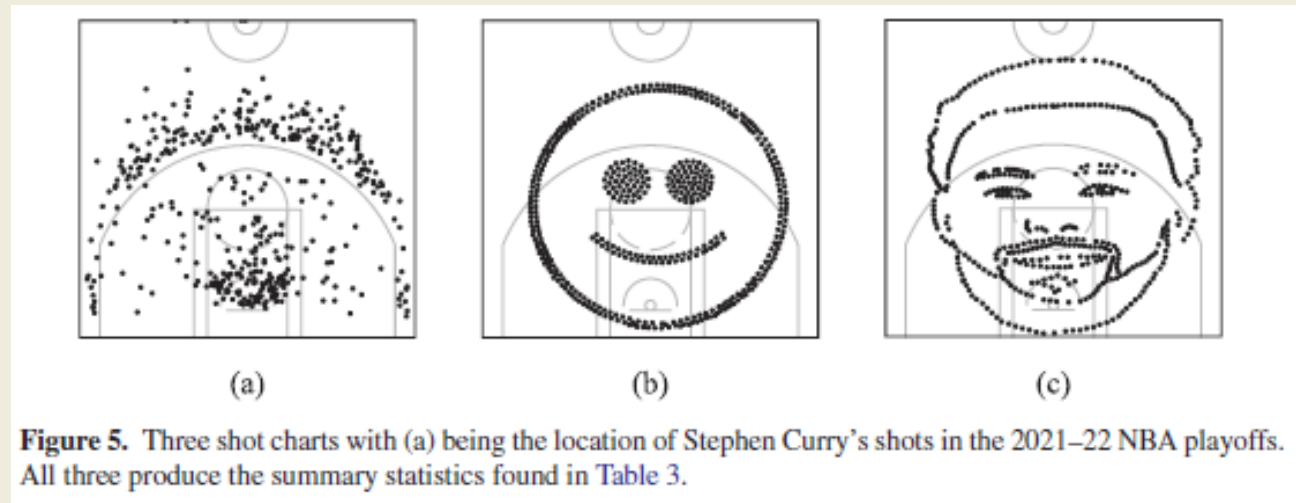
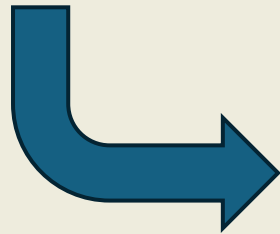
Table 3. Summary statistics for Stephen Curry's shot locations (in feet) in the 2021–22 NBA playoffs.

Anscombe's quartet

Steph Curry's 2021-2022 NBA Playoff shot locations

- Simple summary statistics can obscure actual data distribution (shot location).

mean of x	0.73977
mean of y	14.71046
standard deviation of x	11.26635
standard deviation of y	10.42091
correlation of x and y	-0.00989



Anscombe's quartet

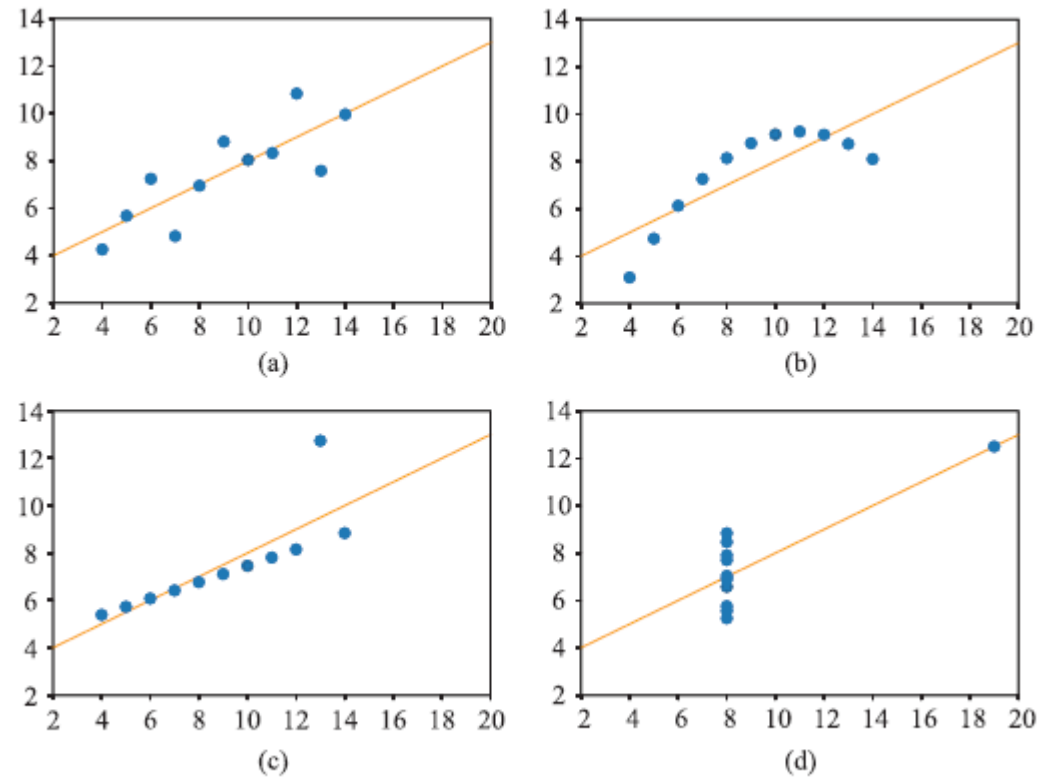


Figure 4. Anscombe's quartet have similar summary statistics but exhibit significant differences when plotted.

In-Class Case Study: Describing Your Project Data

In-Class Case Study

- Before conducting analysis, it is good to find out what your data actually looks like. In this activity, you will describe one important numeric variable using measures of center, spread, and distribution.

In-Class Case Study

Part 1: Choose what to describe

- Identify what **one row** of your dataset represents: a player, a player-game, a team-game, a training session, or something else.
- Choose one numeric variable that matters.
- Record the variable's units and the number of observations with a usable value.

If you do not have data, use a provided class dataset and complete the same steps. Note which dataset you used.

In-Class Case Study

Part 2: Summarize the variable

- Calculate:
 - Mean and median
 - Minimum and maximum
 - Range
 - Standard deviation
- Make one plot that shows the distribution

In-Class Case Study

Part 3: Interpret what you found

- Write three or four sentences that address these questions:
 1. What would you tell someone is a **typical** value? Would you use the mean or median, and why?
 2. How much do observations vary? Use at least one measure of spread in your answer.
 3. What does the distribution show that the summary statistics alone might hide? Consider unusual values, skew, gaps, or clusters.

In-Class Case Study

Bonus:

- Compare the variable across a relevant group, such as position, team, season, or competition level.
 1. Report the number of observations in each group and compare both center and spread.
 2. Describe what you observe without assuming the group difference has a particular cause.